BUSINESS INTELLIGENCE LAB

EXPERIMENT – 1

Data Acquisition and Data Profiling using Microsoft Power BI

Practical Documentation

Item	Details
Software	Microsoft Power BI Desktop
Data Sources	CSV, Excel Workbook and SQL Database
Power Query Features	Column Quality, Column Distribution, Column Profile
Datasets	Sales_Data.csv, Employee_DB.xlsx, Transactions

Prepared for Business Intelligence Lab Submission

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1. AIM

To acquire data from multiple heterogeneous sources and perform data profiling using Microsoft Power BI Desktop in order to understand the structure and quality of data, identify potential data-quality issues, and prepare the datasets for further analysis and visualization.

2. OBJECTIVES

- Understand the role of Business Intelligence in data-driven decision-making.
- Understand the concept of data acquisition from multiple sources.
- Import data from CSV, Excel and SQL database sources into Power BI.
- Use Power Query Editor for data inspection and profiling.
- Analyze data quality using Column Quality, Column Distribution and Column Profile.
- Examine data types, missing values, errors, distinct values and unique values.
- Identify observations that may require attention before further analysis.

3. TOOLS / SOFTWARE REQUIRED

- Microsoft Power BI Desktop
- Power Query Editor
- Sales_Data.csv
- Employee_DB.xlsx
- Transactions / SQL database
- SQL Server / SQL Server Management Studio for the SQL-based source

4. DATASET USED

Three heterogeneous datasets were used in this experiment. They were brought into the same Power BI report so that data acquisition and profiling could be demonstrated in one BI environment.

Dataset	Source Type	Purpose
Sales_Data.csv / Sales_Data (1)	CSV / Text file	Sales-related transactional information
Employee_DB.xlsx / Employees	Excel workbook	Employee information
Transactions	SQL database	Transaction-related information

5. THEORY

5.1 Business Intelligence

Business Intelligence (BI) is the process of collecting, integrating, transforming, analyzing and presenting data to generate meaningful information for decision-making. A typical BI workflow can be represented as Data Sources → Data Acquisition → Data Preparation → Data Analysis → Visualization → Decision Making.

In an organization, information may be distributed across CSV files, spreadsheets and relational databases. A BI platform such as Power BI helps bring these different sources into a common analytical environment.

5.2 Data Acquisition

Data acquisition is the process of collecting or importing data from one or more sources into an analytical environment. Power BI provides the Get Data functionality for acquiring data from sources such as CSV files, Excel workbooks, SQL Server, web services and other databases.

In this experiment, CSV, Excel and SQL sources were used to demonstrate acquisition from heterogeneous data sources.

5.3 Heterogeneous Data Sources

A heterogeneous data environment contains data stored in different formats, structures or systems. CSV is a file-based tabular format, Excel is a spreadsheet-based format, while SQL Server stores data in relational tables. Power BI can connect to all three and expose them as queries for further processing.

5.4 Power Query Editor

Power Query Editor is the data preparation environment in Power BI. It is used to inspect, clean, transform and profile data before it is used for reporting. In this experiment, it was primarily used to inspect the three imported datasets and evaluate their data quality.

5.5 Data Profiling

Data profiling is the systematic examination of a dataset to understand its structure, content and quality before analysis. It helps identify missing values, errors, unexpected data types, duplicate patterns and unusual distributions.

5.6 Data Quality Dimensions

Dimension	Meaning
Completeness	Checks whether required values are present.
Validity	Checks whether values follow the expected format or data type.
Uniqueness	Checks whether identifier values are unique where uniqueness is expected.
Consistency	Checks whether similar information follows a consistent format.
Accuracy	Checks whether values correctly represent the intended real-world information.
Timeliness	Checks whether the data is sufficiently current for the intended analysis.

5.7 Power BI Profiling Features

Column Quality:

Shows the percentage of valid, error and empty values for each column.

Column Distribution:

Shows value distribution along with distinct and unique counts.

Column Profile:

Provides detailed statistics for the selected column, including count, empty/error values and, where applicable, minimum, maximum and average.

6. PROCEDURE

The practical was performed in stages. First, the imported data sources were verified in Power BI Desktop. Next, Power Query Editor was opened and its profiling features were enabled. Finally, each dataset was inspected and the observations were documented.

Step 1 – Launch Power BI Desktop

Microsoft Power BI Desktop was opened and the prepared BI_Experiment_1 report was loaded. The report contained the data required for the experiment.

Purpose: To establish the Power BI environment in which the heterogeneous datasets could be inspected.

Step 2 – Verify the Imported Data Sources

The Data pane on the right side of Power BI Desktop was inspected. Three tables were visible: Employees, Sales_Data (1) and Transactions. Their presence confirmed that the three data sources had been acquired and loaded into the report.

Purpose: To verify successful data acquisition before profiling.

Step 3 – Verify Sales_Data (1)

The Sales_Data (1) table was inspected. The visible fields included SaleID, Date, CustomerID, Product, Quantity, UnitPrice and SalesAmount. The captured profiling view contained 200 rows.

Purpose: To verify the structure of the sales dataset and prepare it for profiling.

Step 4 – Verify Employees

The Employees table was inspected. The visible fields included EmployeeID, Name, Department, Region, Salary and JoiningDate. The captured dataset contained 100 rows and 6 columns.

Purpose: To verify the employee dataset structure.

Step 5 – Verify Transactions

The Transactions table was inspected. The visible fields included TransactionID, CustomerID, TransactionDate, Amount and PaymentMode. The captured dataset contained 150 rows and 5 columns.

Purpose: To verify the SQL-based transaction dataset.

Step 6 – Open Power Query Editor

From the Home tab, Transform data was selected. Power Query Editor opened and the three queries—Sales_Data (1), Employees and Transactions—were visible on the left side.

Purpose: Power Query provides the data inspection and profiling environment used in this experiment.

Step 7 – Enable Column Quality

Inside Power Query Editor, the View tab was used to enable Column Quality. Each column then displayed Valid, Error and Empty percentages.

Purpose: To assess completeness and visible data errors.

Step 8 – Enable Column Distribution

Column Distribution was enabled from the View tab. Distribution charts and distinct/unique counts were displayed for the selected columns.

Purpose: To understand value variety and repetition.

Step 9 – Enable Column Profile

Column Profile was enabled. After selecting a column, the lower section displayed statistics such as Count, Error, Empty, Distinct, Unique, Minimum, Maximum and Average where applicable.

Purpose: To understand the detailed statistical characteristics of the data.

Step 10 – Profile Sales_Data (1)

The Sales_Data (1) query was selected and its profiling results were inspected. SaleID showed 200 distinct and 200 unique values. Date showed 152 distinct and 112 unique values. CustomerID showed approximately 99% valid values with a small empty percentage visible. Product had 5 distinct values, Quantity had 5, UnitPrice had 5 and SalesAmount had 17 distinct values.

Purpose: To identify completeness, uniqueness and distribution characteristics in the sales dataset.

Step 11 – Profile Employees

The Employees query was selected. EmployeeID and Name each showed 100 distinct and 100 unique values. Department and Region each showed 4 distinct values. Salary showed 100 distinct and 100 unique values. JoiningDate showed 93 distinct and 86 unique values. The displayed columns showed 100% valid, 0% error and 0% empty.

Purpose: To assess the quality and structure of employee data.

Step 12 – Profile Transactions

The Transactions query was selected. TransactionID showed 150 distinct and 150 unique values. CustomerID showed 80 distinct and 37 unique values. TransactionDate showed 126 distinct and 105 unique values. Amount showed 136 distinct and 123 unique values. PaymentMode showed 3 distinct and 0 unique values. The displayed columns showed 100% valid, 0% error and 0% empty.

Purpose: To inspect transaction-level completeness, uniqueness and category distribution.

Step 13 – Check Data Quality

The Column Quality results were compared across the datasets. The Employee and Transactions views showed 100% valid, 0% error and 0% empty in the displayed columns. A small completeness concern was visible for CustomerID in Sales_Data (1).

Purpose: To identify fields that may require additional review before analysis.

Step 14 – Check Identifier Uniqueness

Identifier fields were examined using the distinct and unique counts. EmployeeID, SaleID and TransactionID matched their row counts in the captured views, indicating unique identifiers.

Purpose: To confirm that key identifier fields were unique in the captured datasets.

Step 15 – Interpret Repeated Values

Repeated values were interpreted according to the meaning of the field. For example, repeated CustomerID values are normal because one customer may make multiple transactions, and a small number of PaymentMode categories is expected for a categorical field.

Purpose: To distinguish legitimate repetition from actual duplicate records.

Step 16 – Verify Data Types

The data type indicators in Power Query were inspected. Text, whole-number, decimal and date types were visible according to the selected fields. Appropriate data types are important for calculations, filtering, sorting and visualization.

Purpose: To ensure that fields are represented using suitable data types.

Step 17 – Apply the Data

After completing inspection and profiling, Close & Apply was used to return from Power Query Editor to Power BI Desktop. The queries remained available for subsequent analysis and visualization.

Purpose: To complete the data preparation stage and return the profiled data to the report.

7. RESULT AND SCREENSHOT EVIDENCE

The following screenshots were captured from the actual Power BI Desktop practical. They provide visual evidence of the data acquisition and profiling work performed during the experiment.

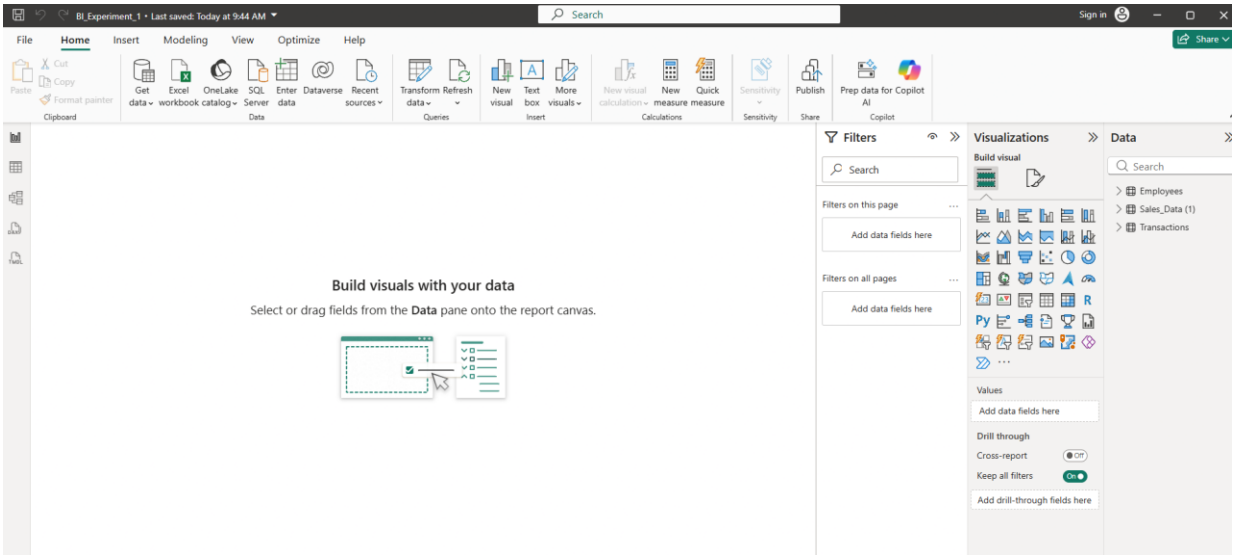


Figure 1: Power BI Desktop showing the three loaded data sources — Employees, Sales_Data (1) and Transactions.

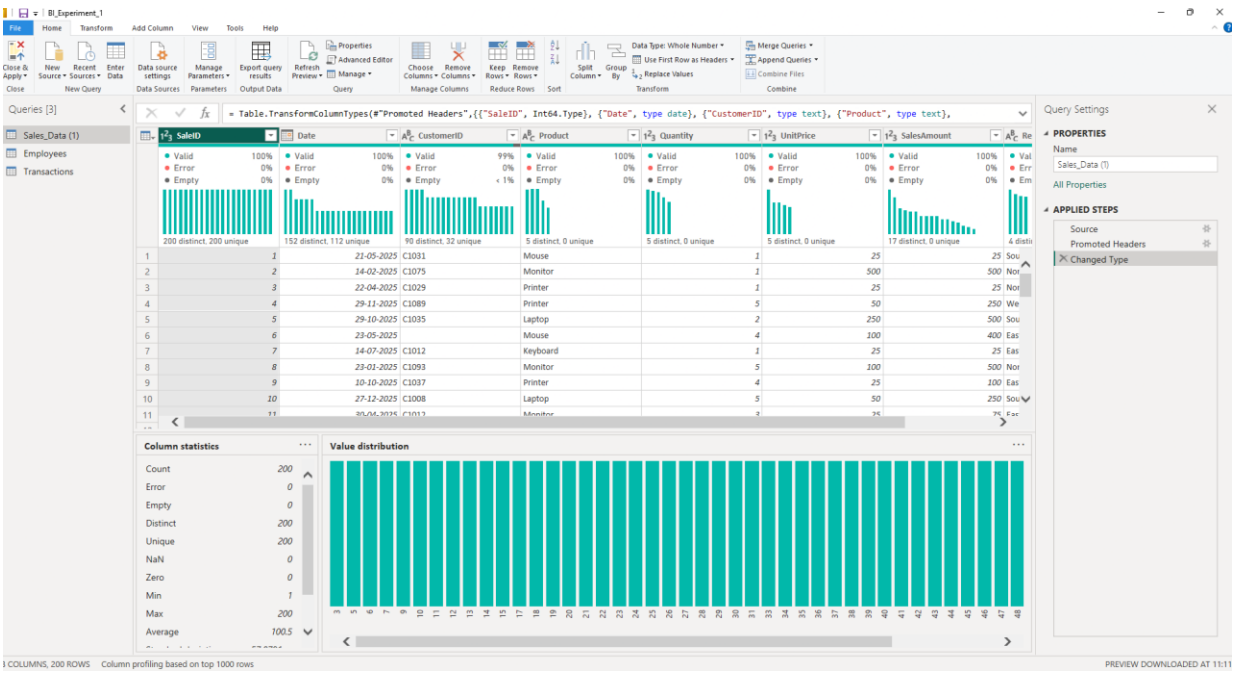


Figure 2: Sales_Data (1) profiling in Power Query using Column Quality, Column Distribution and Column Profile.

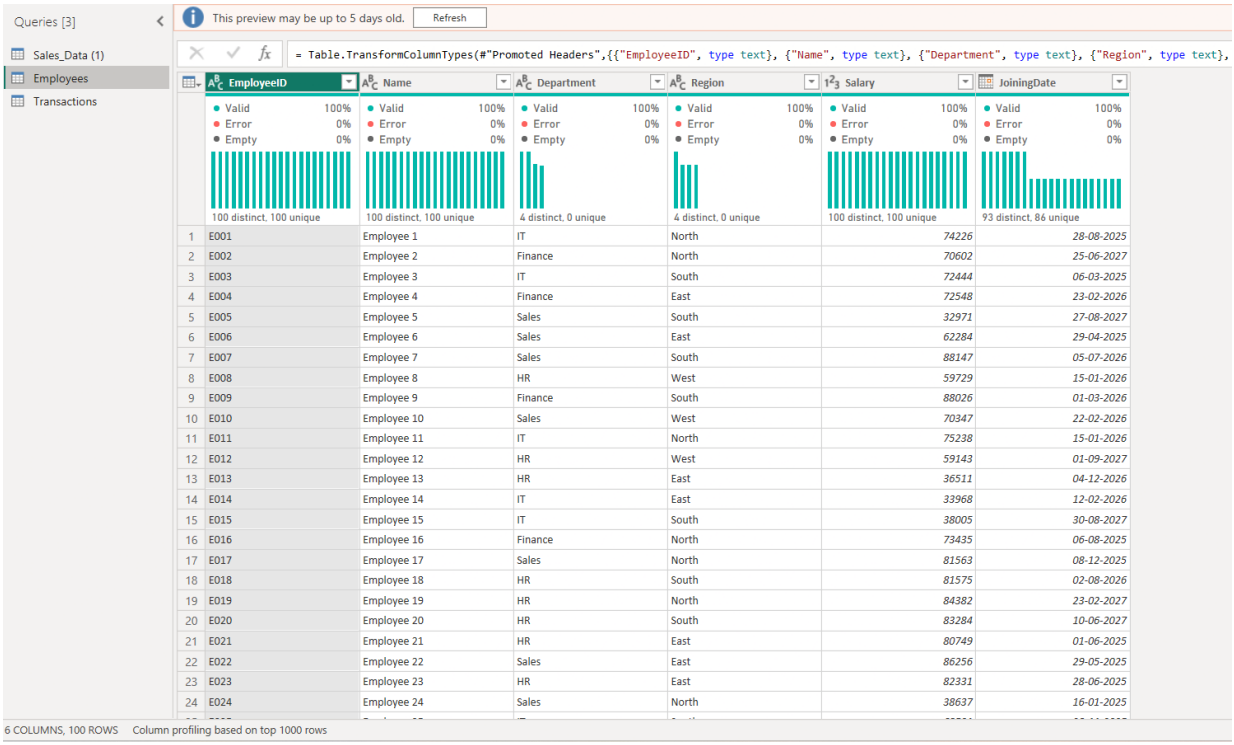


Figure 3: Employees dataset profiling. The displayed columns show 100% valid, 0% error and 0% empty values.

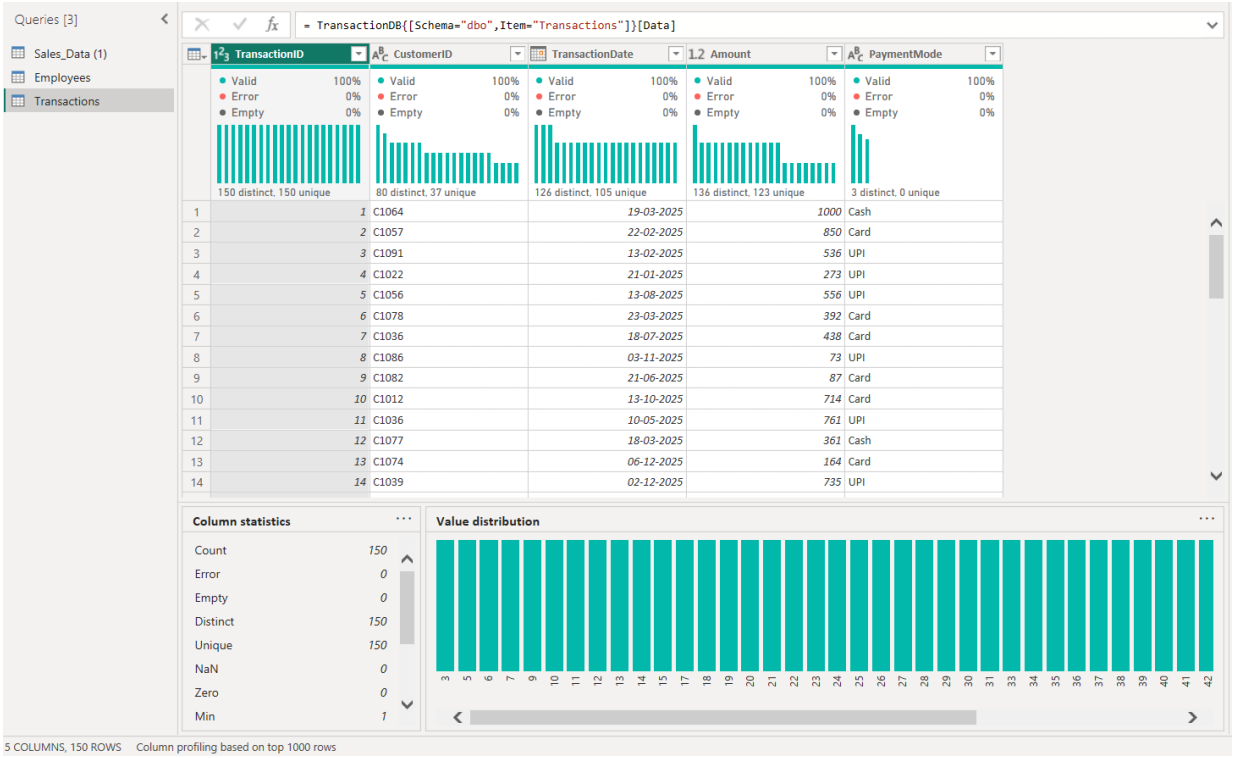


Figure 4: Transactions dataset profiling showing distinct/unique counts and column statistics.

Based on the captured Power Query results, the experiment successfully demonstrated data acquisition from three heterogeneous sources and data profiling using Power BI. The Employee and Transactions profiling screens showed 100% valid values with 0% errors and 0% empty values in the displayed columns. The Sales_Data (1) profiling screen showed a small completeness concern in CustomerID, while the main identifier fields SaleID, EmployeeID and TransactionID were observed to be unique.

8. DATA QUALITY OBSERVATIONS

Dataset / Field	Observation	Interpretation / Recommended Action
Sales_Data (1) – CustomerID	Approximately 99% valid with a small empty percentage visible.	Inspect the rows with missing CustomerID and decide whether they should be corrected, mapped or retained as unknown.
Employees – EmployeeID	100 distinct and 100 unique values.	Identifier is unique in the captured dataset.
Sales_Data (1) – SaleID	200 distinct and 200 unique values.	Identifier is unique in the captured dataset.
Transactions – TransactionID	150 distinct and 150 unique values.	Identifier is unique in the captured dataset.
Transactions – PaymentMode	3 distinct and 0 unique values.	This is normal categorical repetition and should not be treated as an error.
Transactions – CustomerID	80 distinct and 37 unique values.	Repeated customers are expected in transaction data; repetition is not automatically duplication.

9. LEARNING OUTCOMES

- Understood the role of Business Intelligence in data-driven decision-making.
- Learned how to acquire data from heterogeneous sources.
- Learned how to load CSV, Excel and SQL-based data into Power BI.
- Gained practical experience with Power Query Editor.
- Learned how to use Column Quality, Column Distribution and Column Profile.
- Understood the difference between distinct and unique values.
- Learned how to identify missing and error values.
- Learned how to verify the uniqueness of identifier columns.
- Understood why repeated values are not always duplicate records.
- Understood why data profiling should be performed before analysis and visualization.

10. CONCLUSION

This experiment demonstrated the initial data-preparation stage of a Business Intelligence workflow. Data from CSV, Excel and SQL sources was made available in Microsoft Power BI Desktop and inspected through Power Query Editor. The built-in profiling features provided a systematic method for evaluating completeness, validity, uniqueness, distribution and basic statistics. The practical work demonstrated that data profiling should be performed before dashboard creation because reliable BI insights depend on well-understood and appropriately prepared data.